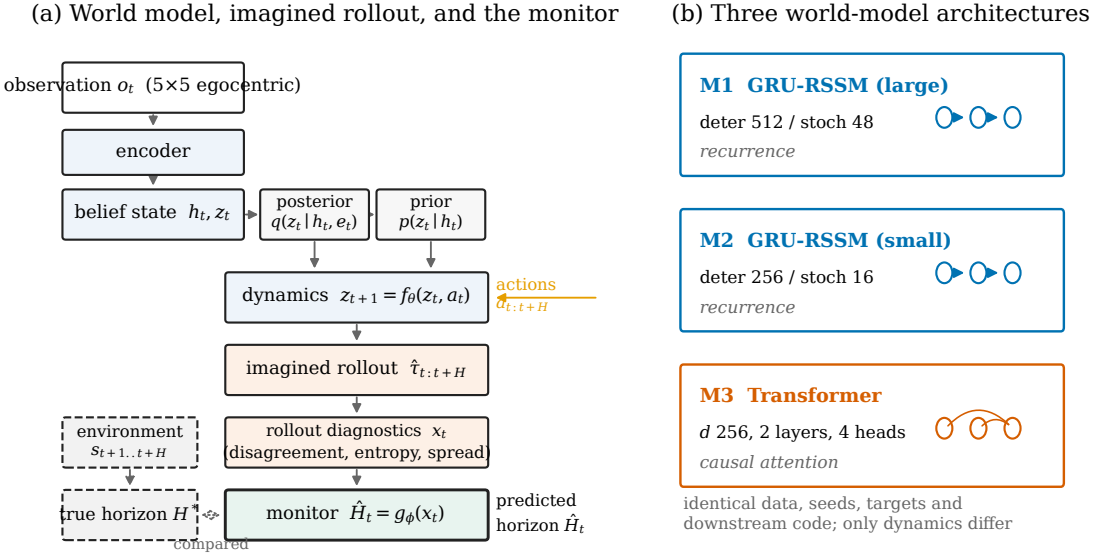
(a) World model, imagined rollout, and the monitor
observation o_t (5×5 egocentric)
encoder
belief state h_t, z_t
posterior $q(z_t \mid h_t, e_t)$
prior $p(z_t \mid h_t)$
dynamics $z_{t+1} = f_\theta(z_t, a_t)$
actions $a_{t:t+H}$
imagined rollout $\hat{\tau}_{t:t+H}$
environment $s_{t+1..t+H}$
rollout diagnostics x_t (disagreement, entropy, spread)
true horizon $H^\star$
compared
monitor $\hat{H}_t = g_\phi(x_t)$
predicted horizon $\hat{H}_t$
(b) Three world-model architectures
M1 GRU-RSSM (large)
deter 512 / stoch 48
recurrence
M2 GRU-RSSM (small)
deter 256 / stoch 16
recurrence
M3 Transformer
d 256, 2 layers, 4 heads
causal attention
identical data, seeds, targets and downstream code; only dynamics differ